

Employee Salary Analysis Report: Gender & Employment Distribution

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01. Introduction to the Dataset

This report provides a statistical analysis of an employee dataset containing records for (474) employees. The analysis focuses on **understanding structure** of the organization's salary, and **identifying differences** based on gender and employment roles.

The dataset includes the following **variables**:

- ➔ Gender (Qualitative): Male or Female.
- ➔ Employment (Qualitative): Clerical, Custodial, or Manager.
- ➔ Current Salary (Quantitative): Annual return in USD.

The dataset's **gender distribution** consists of (258|54.4%) males & (216|45.6%) females.

The **Objective** of my analysis is to evaluate salary distributions and investigate the relationship between gender, job category, and salary.

02. Description of the Analysis Performed

The following statistical methods were applied:

- ➔ Descriptive Statistics: Mean and Median was calculated to determine salary levels.
- ➔ Dispersion Measurements: Standard Deviation, Range, Variance, and IQR were used to determine the spread and variability of salaries.
- ➔ Distribution Analysis: Skewness were examined to understand the shape of the data distribution.
- ➔ Box-plot Analysis: Employed to visually identify outliers and compare distributions.
- ➔ Crosstabulation: analyze the relationship between qualitative variables.

03. Discussion of Findings and Conclusions

3.1. Salary by Gender

	Mean Salary	Median Salary	Standard Deviation
Male	\$41,441.78	\$32,850.00	\$19,499.21
Female	\$26,031.92	\$24,300.00	\$7,558.02

Central Tendency

The table reveals a significant gap in payment between genders.

The **Mean** salary for males is (\$41,441.78), whereas the **Mean** salary for females is (\$26,031.92), This results in a difference of (\$15,409.86).

Skewness Analysis

Both distributions show positive **skewness**, with **skewness** values of (1.863) for females and (1.639) for males.

In both cases, the **mean** is higher than the **median**, indicating that high income earners are pulling the average upward.

Therefore, the **median** serves as a more accurate measure of the "average" salary for this dataset.

Dispersion Analysis

Male salaries demonstrate massive **variance** compared to female salaries.

The **standard deviation** for males is (\$19,499.21) with a range of (\$115,350).

In the other side, female salaries are much more clustered, with a **standard deviation** of (\$7,558.02) and a range of (\$42,375).

The **IQR** confirms this, with the male **IQR** (\$22,675) being triple the female **IQR** (\$7,013).

3.2. Relation Between Employment and Gender

	Female Count	Female %	Male Count	Male %
Clerical	206	95.4%	157	60.9%
Custodial	0	0.0%	27	10.5%
Manager	10	4.6%	74	28.7%
Total	216	100.0%	258	100.0%

A crosstabulation of gender and employment category reveals distinct industrial separation.

Key Observation

There is a clear "Managerial Dominance" by males.

From the 84 Manager positions, (88.1%|74) are held by males, while only (11.9%|10) are held by females.

In opposition, (95.4%) of the female workforce is concentrated in Clerical roles.

3.3. Salary Distribution and Outliers

Visual inspection via **boxplots** confirms the statistical findings regarding dispersion.

The male **boxplot** displays a much wider **IQR** that is positioned significantly higher on the salary axis than the female **boxplot**.

Serious **outliers** are present in the male salary data, reaching up to (\$135,000).

These specific high **outliers** are the primary mathematical cause for the heavy positive **skewness** and the inflated **mean** salary observed in the male side.

3.4. Salary by Employment Category

Analyzing salary by role clarifies the hierarchy of compensation:

- ➔ Manager: Highest return (Mean: \$63,977.80, Median: \$60,500.00) and highest dispersion (Std. Dev: \$18,244.78).
- ➔ Custodial: Moderate return (Mean: \$30,938.89) with the tightest pay structure (Std. Dev: \$2,114.62).
- ➔ Clerical: Lowest return (Mean: \$27,838.54, Median: \$26,550.00).

Conclusion

The data analysis confirms a clear and significant salary gap favoring male employees, who earn approximately (59%) more on average than their female colleagues.

However, the analysis by employment category suggests that this gap is not merely a result of unequal pay for the same work, but rather a result of "business" sorting.

Males dominate the highest-paying category (Managers | 88.1%), while females are heavily concentrated in the lowest-paying category (Clerical | 95.4%).

The gender pay gap in this organization is driven by the distribution of genders across job categories rather than dispersion within the categories themselves.

Due to the positive **skewness** caused by high-earning male managers, **median** salaries remain the most representative measure for typical employee payments.